

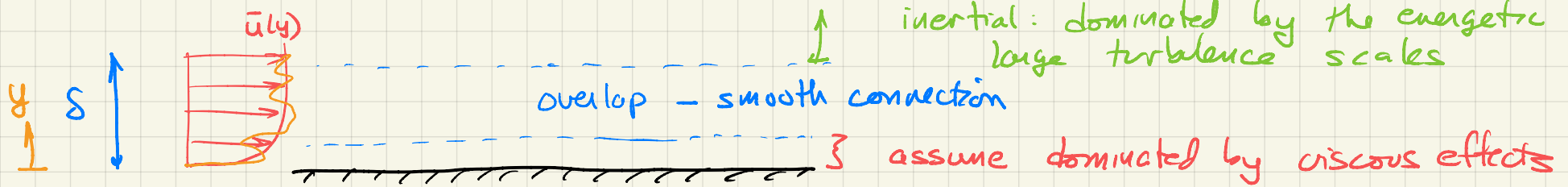
Lecture 31

Turbulence

- Log law
- TKE transport

BL turbulence

Log Law



INNER

$$\bar{u}(y) = f(\tau_w, \rho, \mu, y)$$

↑
not δ

wall shear stress

OUTER

$$U - \bar{u} = g(\tau_w, \rho, \mu, \delta)$$

↑
not μ

non dimensionalize...

$$\frac{1}{5} = f\left(\frac{y}{\delta}\right)$$

$$\frac{U - \bar{u}}{u^*} = \tilde{g}\left(\frac{y}{\delta}\right)$$

$$u^* = \left(\frac{\tau_w}{\rho}\right)^{1/2}$$

mass independent

"friction velocity"

— τ_w converted to units of velocity

— a kinematic "idea" like velocity

OVERLAP:

$$\frac{1}{y} \frac{1}{y^*} = f\left(\frac{y}{y^*}\right) = \frac{C}{y^*} - \tilde{g}\left(\frac{y}{y^*}\right)$$

$\underbrace{\frac{y}{y^*}}_{\text{same}} \times \underbrace{\frac{1}{y^*}}_{\text{same}}$

form

$$f(xy) = C + g(y)$$

→ f, g must be log's

$$A \log x + B \log y = C + B \log y$$

$$\frac{1}{y} \frac{1}{y^*} = \frac{1}{K} \log \frac{y}{y^*} + B$$

constants

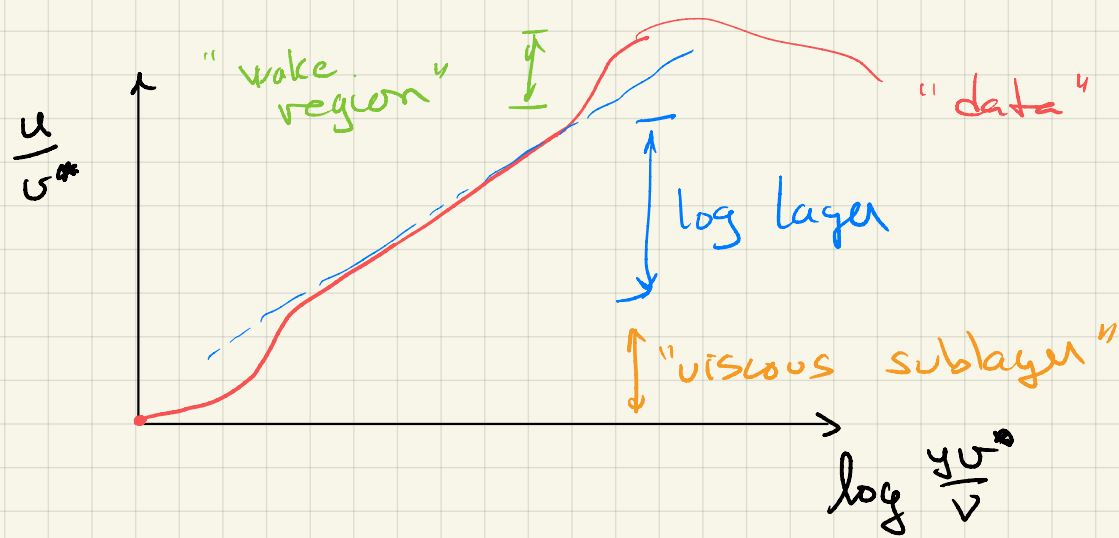
only way to get an independent y term "out of" the f(xy)

$$\frac{1}{y} \frac{1}{y^*} = -\frac{1}{K} \log \frac{y}{y^*} + A$$

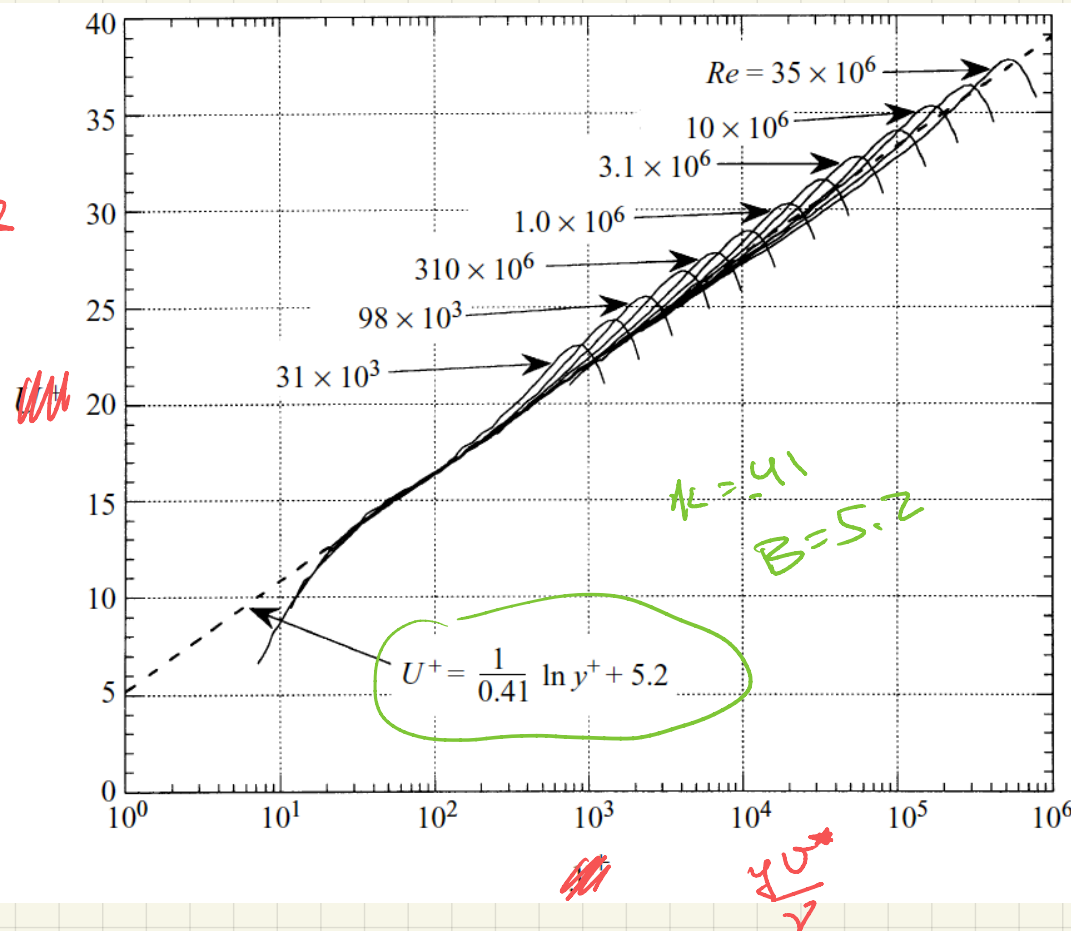
presumed universality

[NOTE: no $\frac{\partial \rho}{\partial x}$ assumed]

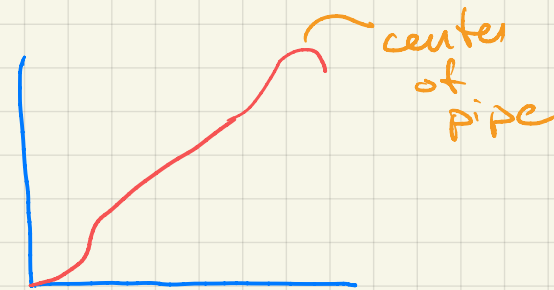
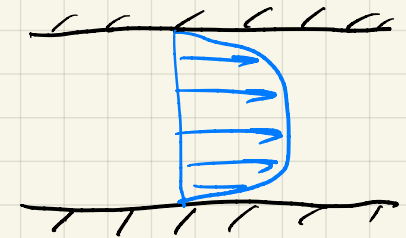
→ only need to become log in overlap



Zargola
Smits
1998



Pipe flow



- Turbulence Kinetic Energy (TKE) in BL

Q: governing eq for $\underbrace{\frac{1}{2} \overline{u'_i u'_i}}_k$ $\xleftarrow{\text{kinetic energy}}$
mass

$\bar{u}$ eq. $\bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial \overline{u'_i u'_j}}{\partial x_j}$

full u
(mom) $\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j \partial x_j}$

subtract
 $\Rightarrow u'_i$ $\frac{\partial u'_i}{\partial t} + \frac{\partial}{\partial x_j} (u_j u_i - \bar{u}_j \bar{u}_i) = -\frac{1}{\rho} \frac{\partial p - \bar{p}}{\partial x_i} + \nu \frac{\partial^2 (u_i - \bar{u}_i)}{\partial x_j \partial x_j} + \frac{\partial \overline{u'_i u'_j}}{\partial x_j}$

$\times u'_i \quad \dots \quad \text{average} \quad \dots$

$$\underbrace{\frac{\partial K}{\partial t} + \bar{u}_j \frac{\partial K}{\partial x_j}}_{\substack{\text{advection by} \\ \text{mean flow} \\ \bar{u} \\ \frac{D K}{D t}}} = - \underbrace{\frac{1}{\rho} \frac{\partial \overline{p' u_i'}}{\partial x_j}}_{\substack{\text{"pressure diffusion"} \\ \nabla \cdot (\bar{p} \bar{u}) \\ \text{transport}}} - \underbrace{\frac{1}{2} \frac{\partial \overline{u_i' u_i' u_j'}}{\partial x_j}}_{\substack{\nabla \cdot [\bar{u} (\bar{u} \cdot \bar{u})] \\ \text{transport by turbulence fluctuation}}} + \underbrace{\nu \frac{\partial^2 K}{\partial x_j \partial x_j}}_{\substack{\nabla^2 K \\ \text{"viscous diffusion"} \\ \text{spreads the TKE via viscosity}}} - \underbrace{\overline{u_i' u_j'} \frac{\partial \bar{u}_i}{\partial x_j}}_{\substack{\text{"production"} \\ \text{Reynolds stress interacting with the mean flow}}} - \underbrace{\nu \frac{\partial u_i'}{\partial x_j} \frac{\partial u_i'}{\partial x_j}}_{\substack{\geq 0 \\ \varepsilon \\ \text{"viscous dissipation"}}}$$

$\downarrow$ fluctuations moving fluctuation energy
 $\uparrow$ div form: $\nabla \cdot (\nu \nabla K)$ not source or sink

$\downarrow$ shows up with opposite sign in mean KE equation

The point?

→ provides "scales" for turbulence closures

$$\overline{u' u'} = \underline{\underline{f}}(\bar{u}) \quad \text{the closure problem}$$

$\uparrow$ easier if K included

